

ANNEXURE A: PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Question 1: P2P File Sharing Algorithm

Topic: P2P Communication

Scenario:

A student wants to download a file from another computer in a peer-to-peer network. The system must connect to an available peer, check whether the peer has the requested file, download the file if available, display a failure message if the file is unavailable, and end the process.

Correct Algorithm Sequence:

1. START
2. CONNECT TO PEER
3. IF FILE IS AVAILABLE THEN
4. DOWNLOAD FILE
5. ENDIF
6. DISPLAY "File not available"
7. END

Question 2: VoIP Call Connection Algorithm

Topic: VoIP Communication

Scenario:

A user starts a voice call using a VoIP application. The application checks whether an internet connection is available. If the internet is available, it connects the call and starts voice communication. Otherwise, it displays an error message and ends the process.

Correct Algorithm Sequence:

1. START
2. CHECK INTERNET CONNECTION
3. IF INTERNET IS AVAILABLE THEN
4. CONNECT CALL
5. START VOICE COMMUNICATION
6. ELSE
7. DISPLAY "No Internet Connection"
8. ENDIF
9. END

This correctly handles both conditions:

- Internet available → connect call → start voice communication
- No internet → display error

Question 3: Overlay Network Message Routing Algorithm

Topic: Overlay Networks

Scenario:

A user wants to send a message to another node in an overlay network. The system checks whether the destination node is available. If the node is available, the message is sent directly. Otherwise, the message is forwarded through another available node. The system then confirms that the message has been sent.

Correct Algorithm Sequence:

1. START
2. CHECK DESTINATION NODE
3. IF DESTINATION NODE IS AVAILABLE THEN
4. SEND MESSAGE DIRECTLY
5. ELSE
6. FORWARD MESSAGE THROUGH ANOTHER NODE
7. DISPLAY "MESSAGE SENT"
8. ENDIF
9. END

This gives:

- Destination available → send directly
- Destination unavailable → forward through another node
- Then → confirm message sent → end.